Engineering Mathematics - I · Unit_01 · Unit 1

DRAFT — UE26MA141A · Unit_01 · Unit 1 · Problems on Euler's theorem

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Problems on Euler's theorem

1 Problems on Euler's theorem

Study focus: Problems on Euler's theorem. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000146 · EV0000147

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we will solve problems on homogeneous functions and Euler's theorem.

Key details

- Remember, here n can be positive, negative.
- Or even zero.
- Then we are left out with x power 4 plus 2x square y square plus y power 4.
- In the denominator we get lambda power 4 of x square plus y square whole square.
- After lambda power 4, lambda power 4, we can write 1 to be lambda power 0.
- U equals sine inverse of y by x divided by square root of x square plus y square.
- Replace x by lambda x, replace y by lambda y.
- Sign in us of lambda y by lambda x.
- And in the denominate denominator, it is square root of lambda x whole square plus lambda y whole square.
- U equals x square y plus xy square divided by x plus y power 3.
- U equals e power x minus e power minus y divided by x minus y.
- This simplification doesn't give us lambda power n times u of x comma y.
- It means u of lambda x comma lambda y is not equal to lambda power n times u of x comma y.

Example or application from the lecture

- Let us solve the first problem u equals x power four plus two times x square into y square plus y power four divided by x square plus y square one square.
- Now in order to test whether the given function is homogeneous or not, we need to prove this particular condition that is u of lambda x comma lambda y equals lambda power n times u

of x comma y.
- In the given function, we replace x by lambda x, y by lambda y.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000146 · EV0000147 · EV0000148 · +1 source units in pack.json

Core explanation · 05:00–09:59

I hope the concept of homogeneous function is clear to you.

Key details

- If u equal to x minus y divided by x square plus y square, then find the value of x into dou u by dou x.
- plus y into dou u by dou y and x square into dou square u by dou x square plus 2 x y into dou square u by dou x into dou y plus y square into dou square u by dou y square.
- As we know that this is nothing but extension of Euler's theorem.
- So in order to use Euler's theorem, first we need to prove that u is a homogeneous function of degree n.
- We get to know that u is a homogeneous function of degree minus 1 because u of lambda x comma lambda y equals lambda power minus 1 times u of x comma y.
- After proving u to be homogeneous function, we can directly use Euler's formula, which is nothing but x into dou by dou x plus y into dou by dou y equals n u.
- Because n is minus 1, we can get this particular answer that is x into dou by dou x plus y into dou by dou y equals minus u becomes minus half x minus y divided by x square plus y square.
- Because n is minus 1, finally we get 2 times.
- X minus y divided by x square plus y square.
- U of x comma y becomes log u, then this part becomes x power 3 times y power 3 divided by x square plus y square.
- Then check whether f of u is homogeneous or not.
- After we simplify, we get lambda power x power 3 y power 3 divided by lambda square of x square plus y square.
- equals n times f of u by f dash of u.
- If u equal to log of x square plus y square divided by x plus y, then find a value of these two.
- Again observe carefully.
- Again, we use a different approach because log is involved, we take exponential on both the sides.

Example or application from the lecture

- After understanding the concept of homogeneous functions, let us solve problems on Euler's theorem.
- It is given that u equals e power x power 3 times y power 3 divided by x square plus y square, then find the value of x into dou u by dou x plus y into dou u by dou y using euler's theorem.
- Observe carefully, the given function is not homogeneous.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000150 · EV0000151 · EV0000152 · +1 source units in pack.json

Core explanation · 10:00–14:59

Then that leads to f of u equals a power u equals x square plus y square divided by x plus y.

Key details

- Then check whether f of u is homogeneous or not.
- or degree n equals to 1.
- X square dou square u by dou x square plus 2xy dou square u by doux dou y.
- Plus y square dou square u by dou y square equals g of u of g dash of u minus one.
- If u equal to sine inverse of x plus y divided by square root of x plus square root of y, then find the value of these two.
- After using simplification, we get f of u equals lambda power half times sin u.
- Plus one by two.
- In a similar manner, when we use the extension of Euler's theorem, we get this particular step.
- The right hand side is going to be half times tanue.
- And this part is going to be the derivative of half tan u, that is nothing but half second square u, and then we have minus one.

Example or application from the lecture

- Let us look at this particular problem.
- Again observe carefully, the given function is not homogeneous.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000154 · EV0000155 · EV0000156 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Problems on Euler's theorem transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/006_Problems_on_Euler_s_theorem__0d726228252093d7.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Euler's theorem transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/006_Problems_on_Euler_s_theorem__0d726228252093d7.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Euler's theorem transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/006_Problems_on_Euler_s_theorem__0d726228252093d7.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9985; only the JSON segmentation supplies the displayed timestamps.
Problems on Euler's theorem transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/006_Problems_on_Euler_s_theorem__0d726228252093d7.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.

5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00038	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0048. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00039	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0048. Sidecar variants compare...
SRC00040	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0048. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00041	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0048. Sidecar variants compare...

C Revision and retrieval practice